5.5 Proving Triangle Congruence by SSS

Name _____ Period ____ Date _____

Geometry 5.5 · Student sheet

I can…

- use SSS to decide whether two triangles are congruent.
- use HL for right triangles.
- write a proof that uses SSS.

Vocabulary · fill in the missing word

Legs (of a right triangle) — _____

Hypotenuse — the side _____ the right angle

Part 1 · Discovery: can three sides wiggle?

Build a triangle with strips of 5, 7, and 9 cm. Try to change its shape. Compare with a neighbor.

Shape	Could it change shape?	Matched my neighbor's?
Triangle (3 strips)		
Quadrilateral (4 strips)		

Three sides decide a triangle's _____.

Part 2 · Notes and examples (fill in each note, then work the example)

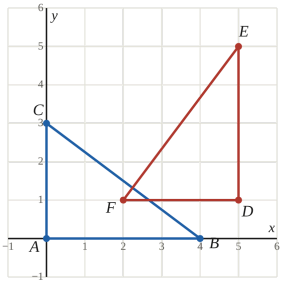
SSS Congruence Theorem: if three _____ of one triangle are congruent to three sides of a second triangle, then the triangles are congruent.

Example 1

One triangle has sides 5, 7, and 9. Another has sides 9, 5, and 7. Are they congruent? Why?

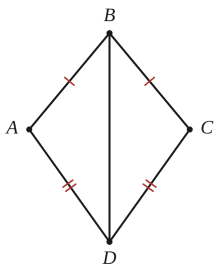
Example 2

△ABC has A(0, 0), B(4, 0), C(0, 3). △DEF has D(5, 1), E(5, 5), F(2, 1). Are they congruent?



Example 3

Given: $\overline{AB} \cong \overline{CB}$ and $\overline{AD} \cong \overline{CD}$. Prove: △ABD ≅ △CBD.



Hypotenuse-Leg (HL) Congruence Theorem: if the hypotenuse and a _____ of a right triangle are congruent to the hypotenuse and a leg of a second right triangle, then the triangles are congruent.

Example 4

Two right triangles each have a hypotenuse of 13 and one leg of 5. Are they congruent? Why?

Example 5

Why do bookshelves and bridges use triangle-shaped braces?

Part 3 · Partner practice

Partner 1

Are triangles with sides 6, 8, 11 and 8, 11, 6 congruent? What about 6, 8, 11 and 6, 8, 12?

Partner 2

Two right triangles each have a hypotenuse of 10 and a leg of 6. Are they congruent? Why?